Lab 1 — Finding Similar Things

Friday, October 16 (~20 min)

Last updated: October 12, 2026.

In class Friday, October 16 (same day as the finish of Lecture 20 / Quiz 6). Quiz 6 is Week 7 matrices / Ax — **not** this lab.

Credit: Lab points are for effort in class, not for a Canvas file upload. Work in Octave or Python on a laptop. Preview the one-pager; write on the hard copy handed out in class.

Pick **Octave or Python** (same choice as Lab 0).

Learning objectives

1. Print vectors and report dimensions/length.
2. Compute distances between feature vectors.
3. Name a nearest neighbor and interpret it in context.

Math connections

Norm and distance (Lecture 07); nearest neighbors (Lecture 09).

Key syntax (this lab)

Task	Octave	Python
Length / shape	`length(x)`, `size(A)`	`len(x)`, `x.shape`
Print	`x` or `disp(x)`	`print(x)`
Subtract	`x-y`	`x-y`
Euclidean norm	`norm(x)`	`np.linalg.norm(x)`

Resources: Lachniet Ch. 1 · Syntax · Resources.

Ops strip (2–3 min)

Three song feature vectors (loudness, tempo) — units already scaled for today:

$$a = (1, 2), \quad b = (2, 1), \quad c = (4, 3).$$

C1.0. Enter `a`, `b`, `c`. Print each and the length of `a`.

Expected: length 2.

Application (~15 min)

C1.1. Compute $\|a - b\|$ and $\|a - c\|$ (Euclidean).

Expected: $\|a - b\| = \sqrt{2} \approx 1.414$; $\|a - c\| = \sqrt{18} = 3\sqrt{2} \approx 4.243$.

Octave: `norm(a-b)`
Python: `np.linalg.norm(a-b)`

C1.2. Which of b or c is the nearest neighbor of query a? One sentence: what does that mean if these were song features?

Expected: b is nearer; among these two songs, b is more similar to a in (loudness, tempo).

C1.3 (scale warning). Replace c by $c_{\text{bad}} = (4, 30)$ (tempo mistyped $\times 10$). Recompute $\|a - c_{\text{bad}}\|$. Does the nearest neighbor of a among $\{b, c_{\text{bad}}\}$ change?

Expected: distance becomes large (≈ 28.16); NN is still b, but the gap is dominated by the bad scale — feature units matter.

One-pager

Preview the Lab 1 ticket; write on the hard copy in class.